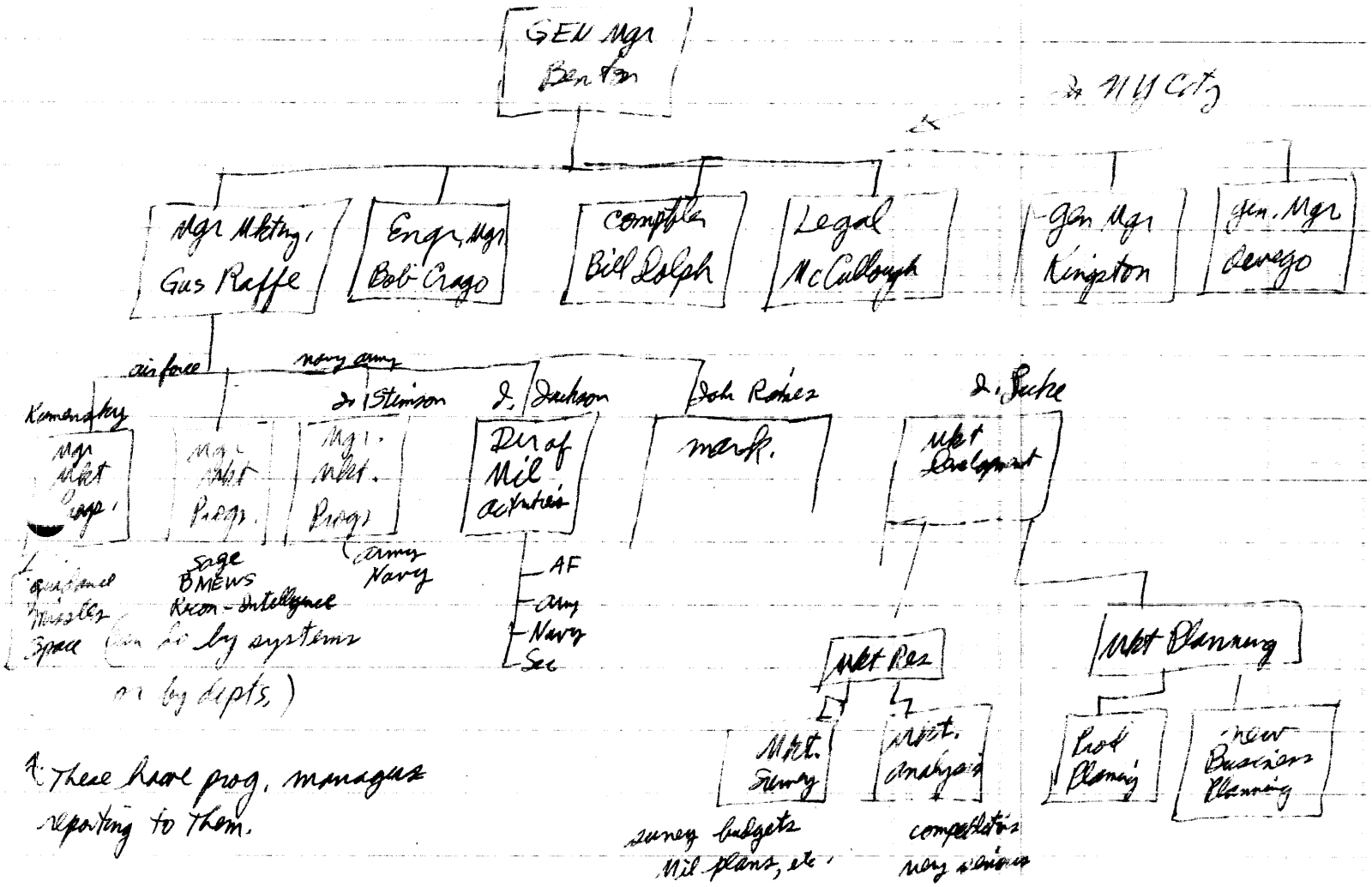
John Lake: Military Products. Aug 15, 58

Structure of MPD



1: These Resrc prog. managers reporting to them.

(W.W. Weather) : Mil spends about $4 billion per year

MPhone 5% of mil business would like 10% - selected

Not trying to get a gen. prod to sell to Mil.

Simple: BMEWS : RCA - Sylvania get contract instead of IBM - GE

who ordered 709's!

ATA being sold to { Q78 36bit wd.
SAC 48bit
Navy

→ question of generality : a military computer for all mil. projects.

Mission: Handle complex special purpose systems for military operational or logistical purposes and/or the aeronautic ~~aeronautic~~ aeronautic fields.

e.g. air traffic control

Motives

Profit motive — a "clean" profit — return on investment (The percentages are equiv to DP because of govt. financing.)

Patriotism —

Res. & Devel. — new knowledge.

give new I/O to DP
new computer
etc.

Products:

1. Systems Management e.g. ~~BMEWS~~ BMEWS
2. Computer Subsystems Q7 etc
3. I/O Systems real time, etc
4. Data Transfer sub-systems. switching systems mic buses, etc (most inadequate in this field)
5. Applied Research e.g. cryogenics, thin films
6. System Studies e.g. air traffic, sonar patterns

M-4-60 Sperry Rand
Moby Dick Sylvania } competing machines.

Airborne systems - Onego

AN/ASQ-38 in B-52's

AN/ASB-4

IBM's systems analog

astro. [redacted]
camera
true heading, } sub contractors.

AN/ASQ-38 B 70 system (classified)

Digital Bomb Nav

DYNABOC 512 word core 12K Mag drum 22 bits + sign 16 bit instr.
light weight etc. miniaturization,
e.g. 10 lb. printer

Circuitry Work at Kingston

R.T.A. (15 Million has been paid to date by contracts)

Speed: 1 μsec mem (cryo) (780,000 instr/sec 1.28 μs/op.
STRETCH. 2.4 μsec mem. (STRETCH) (400,000 instr/sec 2.5 μs/op

Design Features.

- pulse circuit design for low power
- liquid cooled
- Stress diode logic
- 30:1 volume reduction over SAGE

Proposed for SAGE II system. Some kind SAC 48 bit, some UFL, etc.

Pulse Core Logic. (register + matrix logic)

Speed: 30,000 instrs/sec. 33.⅓ μs/op

few transistors

50% less volume than RTA

No cooling

not compatible with RTA circuits.

Proposed for: Management central

Informer – info retrieval.

BASIC PAC
LOGICPAC } system bid.

Adv. SAGE

Speed: 500,000 instr per sec. 2 μs/op

mem: stretch mem

aux drum 210,000 wds.

36 bits

7X more powerful than FSQ-7

¼ floor space of FSQ-7

I/O drum 130,000 wds.

SAC proposed. (similar to 709's)

Army Mobile Computer Prog.

Modules: Basic unit 10,000. instr/sec — 8 μs core mem 4K each

accelerator 30,000. — informer drum

super accelerator 50,000. — I/O converter

complete error checking — Crosstell Unit — communication

1/10 μs 10^{-4}

10,000.

Transistorized Drum

1000 Tracks — 48 bit wd
8K bits/track — 20 fields
8M bits/drum

Max rate 400,000 wds/sec (bit rate per track)

floating heads.
10 year life.

Cryogenic Mem:

Speed 1 μsec or less
capacity 64×64 Lower speed if larg.
Volume 1/100 of ferrite array

Progress 8×8×21 operating in Jan.
1959 in RTA.

Thin Film Mag. Mem.

Speed 2 μsec
1 μsec ultimate

Capacity 64×64

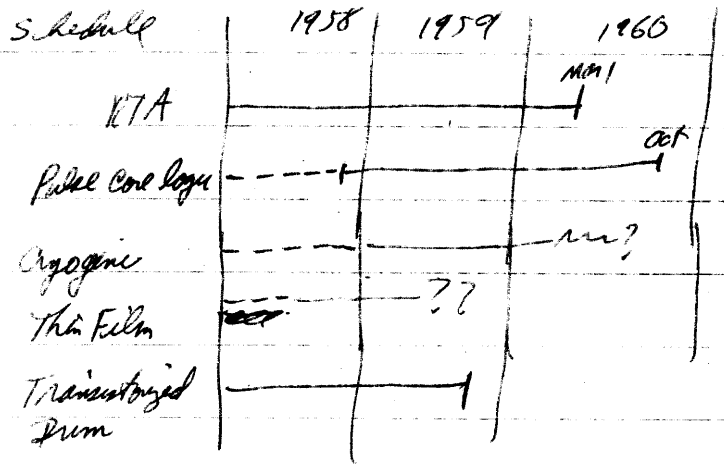
No cooling
low current
Low signal output (1-10 millivolts)

11×16 matrix due Oct 58

Display ECPX

upgrade present equipment

- bright Tubes
- audible annunciation.



		Market
438L	Intelligence	~$2B
465L	SAC Control	$170M
456L	Communications World Wide	$10M
423L	Comm't	~50M
433C	World wide Weather	~50M
117L	Reconnaissance Satellite	~30M

should be compatible.